

Program : Diploma in Chemical Engineering	
Course Code : 6072D	Course Title: Disaster Management (CH)
Semester : 6	Credits: 4
Course Category: Open Elective	
Periods per week: 4 (L: 3 T: 1 P: 0)	Periods per semester: 60

Course Objectives:

- To enable the students to identify the Hazards and Disasters occurring and to familiarize their mitigation measures and the related legislation.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Industrial Mgt & Safety		Industrial Mgt & Safety	5

Course Outcomes :

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Summarize various types of natural and man-made disasters.	12	Understanding
CO2	Summarize the management methods for some of the disasters.	18	Understanding
CO3	Summarize the disaster management in India.	18	Understanding
CO4	Summarize the technological tools for disaster management.	12	Understanding

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1					3		
CO2					3		
CO3					3		
CO4					3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	Summarize various types of natural and man-made disasters.		
M1.01	Paraphrase Disaster, Hazard, Vulnerability, Risk, Capacity.	4	Understanding.
M1.02	Summarize the importance of risk management, assessment and evaluation.	4	Understanding.
M1.03	Summarize the classification of disasters, causes and consequences.	2	Understanding.
M1.04	Summarize the concept of Global Disaster Trends.	2	Understanding.
Contents: Disaster- Concepts and definitions of Disaster, Hazard, Vulnerability, Risk, Capacity and inter-relationship between them. Factors influencing vulnerability and risk. Risk management, assessment and evaluation. Classification - Geological Disasters (earthquakes, landslides, tsunamis, mining); Hydro-Meteorological Disasters (floods, cyclones, lightning, thunder-storms, hail storms, avalanches, droughts, cold and heat waves), Biological Disasters (epidemics, pest attacks, forest fire); Technological Disasters (chemical, industrial, radiological, nuclear) and Manmade Disasters (building collapse, rural and urban fire, road and rail accidents, nuclear, radiological, chemicals and biological disasters) – Case studies of each type of disaster. Global Disaster Trends – Emerging Risks of Disasters – Climate Change and Urban Disasters.			
CO2	Summarize the management methods for disasters.		
M2.01	Summarise the importance of Disaster Management Cycle.	3	Understanding

M2.02	Illustrate the Pre-Disaster management cycle	5	Understanding
M2.03	Illustrate the Disaster management	5	Understanding
M2.04	Illustrate the Post-disaster management cycle.	5	Understanding
	Series Test – I		

Contents:

Disaster Management Cycle – Paradigm Shift in Disaster Management.

Pre-Disaster – Risk Assessment and Analysis, Risk Mapping, zonation and Microzonation, Prevention and Mitigation of Disasters, Early Warning System; Preparedness, Capacity Development; Awareness.

During Disaster – Evacuation – Disaster Communication – Search and Rescue – Emergency Operation Centre – Incident Command System – Relief and Rehabilitation –

Post-disaster – Damage and Needs Assessment, Restoration of Critical Infrastructure – Early

Recovery – Reconstruction and Redevelopment; IDNDR, Yokohama Strategy, Hyogo Framework of Action.

CO3	Summarize the disaster management in India.		
M3.01	Summarise the Disaster Profile of India.	5	Understanding
M3.02	Analyze the major act on Disaster Management	5	Understanding
M3.03	Explain the national Policy on Disaster Management.	5	Understanding
M3.04	Summarise the role of government in the disaster management	3	Understanding

Contents:

Disaster Profile of India – Mega Disasters of India and Lessons Learnt (regional profile and seasonal profile). Hazard zonation map.

Disaster Management Act 2005 – Institutional ((local, state and national) and Financial Mechanism. National Policy on Disaster Management, objectives, National Guidelines and on Disaster

Management, disaster prevention, mitigation and preparedness.

Role of Government (local, state and national), Non-Government and Inter Governmental Agencies. Identification, national response frame work, emergency management, emergency response team, communications.

CO4	Summarize the technological tools for disaster management.		
M4.01	List the roles of Geo-informatics in Disaster Management.	3	Understanding

M4.02	Enumerate the importance of Disaster Communication System	3	Understanding
M4.03	Summarise the importance of Land use Planning and Development Regulations.	3	Understanding
M4.04	Identify the S&T Institutions for Disaster Management in India.	3	Understanding
	Series Test – II		

Contents:

Geo-informatics in Disaster Management (RS, GIS, GPS and RS).

Disaster Communication System (Early Warning and Its Dissemination).

Land Use Planning and Development Regulations, Disaster Safe Designs and Constructions, Structural and Non Structural Mitigation of Disasters

S&T Institutions for Disaster Management in India (NDRF, NDMA, NIDM, ISDR)

Text / Reference

T/R	Book Title/Author
T1	Disaster Management, Rawat Publication; R.B. Singh (Ed)
R1	Publications of National Disaster Management Authority (NDMA) on Various Templates and Guidelines for Disaster Management
R2	An overview on natural & man-made disasters and their reduction; Bhandani, R. K
R3	Management of Natural Disasters in developing countries; Srivastava, H. N., and Gupta G. D
R4	Disaster Management; Ghosh, G. K

Online Resources

Sl.No	Website Link
1	https://onlinecourses.swayam2.ac.in/cec19_hs20/